

Technology #04: Passive Drainage Priority Safety Floor

Technology Domain: Hydraulic Passive Safety + Multi-Lumen Valve Mechanics

Mechanism: CSF shunt with passive safety floor: a secondary drainage path with lower conductance that maintains minimum drainage when the primary path obstructs. Differential pressure thresholds activate the floor path.

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: Common-cause obstruction affects both primary and floor lumens equally (no differential immunity)

Next Experiment: Multi-lumen extruded samples (various $d_{\text{floor}}/d_{\text{primary}}$ ratios)

Buyer Receives: Passive floor concept + mechanism description; Parallel conductance governing equation; Bench test protocol (obstruction, kink, Cpk); Critical common-cause obstruction UNKNOWN disclosure; This engineering dossier with failure analysis + build plan

Buyer Must Build: Production multi-lumen catheter design (dimensioned, toleranced); Multi-lumen extrusion process + vendor qualification; Common-cause obstruction validation evidence; Regulatory submission (510(k) or De Novo); Clinical validation (revision rate endpoint); Manufacturing scale-up

Transfer Posture: Sponsored Validation

Target Buyer: Neuroshunt manufacturer; catheter engineering company

CONFIDENTIAL. See full engineering dossier for details.